# Inverse Functions

Questions 1–39

**Question 1.** For each of the following, state whether or not $f^{-1}$ exists:

a) $f(x) = x^3 + 5$ for $x \in \mathbb{R}$

b) $f : x \mapsto \cos x$ for $x \in \mathbb{R}$

c) $f(x) = \tan x$ for $0 \leq x < \pi,\ x \neq \frac{\pi}{2}$

d) $f(x) = x^2 + 5$ for $x \in \mathbb{R}$

e) $f : x \mapsto x^2 + 4x$ for $x \geq -2$

**Question 2.** Find the inverse of the following functions:

a) $f(x) = 1 - 2x, \quad x \in \mathbb{R}$

b) $f(x) = \frac{x+3}{4}, \quad x \in \mathbb{R}$

c) $f(x) = \frac{7}{2x}, \quad x \in \mathbb{R},\ x \neq 0$

d) $f(x) = \frac{2x+1}{x-3}, \quad x \in \mathbb{R},\ x \neq 3$

e) $f(x) = \frac{4x-9}{2x+1}, \quad x \in \mathbb{R},\ x \neq -\frac{1}{2}$

**Question 3.** Given

$$f : x \mapsto \frac{1}{2}x - 5$$

for $x \in \mathbb{R}$.

a) find $y = f^{-1}(x)$

b) sketch $y = f(x)$ and $y = f^{-1}(x)$ on the same axes

c) Find $f(12)$ and $f^{-1}(1)$, comment on your result.

**Question 4.** For any given one-to-one function, $f$, state the transformation which maps the graph of $y = f(x)$ onto the graph of its inverse function $y = f^{-1}(x)$.

**Question 5.** It is given that the graph of the one-to-one function $y = f(x)$ passes through the given point. State the coordinates of the point corresponding to the same point on the curve $y = f^{-1}(x)$.

a) $(1, 3)$

b) $(-5, 4)$

c) $(3, -3e^2)$

d) $(a, b)$

**Question 6.** For each of the following functions where $x \geq 0$, solve the equation $f(x) = f^{-1}(x)$.

a) $f(x) = 5x - 2$

b) $f(x) = -x + 4$

c) $f(x) = x^2 - 6$

d) $f(x) = -2x^2 - x + 4$

e) $f(x) = x^3$

f) $f(x) = \frac{2}{x+1}$

**Question 7.** Given $f(x) = x^2 + 2x + 5$, explain why $f$ has no inverse.

**Question 8.** Given

$$f : x \mapsto -x + 12, \quad x \neq 0$$

a) show that $ff(x) = x$

b) Hence write down $f^{-1}(x)$

**Question 9.** Given

$$f(x) = \frac{1}{x}, \quad x \neq 0$$

a) show that $ff(x) = x$

b) Hence write down $f^{-1}(x)$

**Question 10.** Given

$$f(x) = \frac{7x - 48}{x - 7}, \quad x \neq 7$$

a) show that $ff(x) = x$

b) Hence write down $f^{-1}(x)$

**Question 11.** Given

$$f(x) = \frac{ax - 1}{x - a}, \quad a > 1$$

a) show that $ff(x) = x$

b) Hence write down $f^{-1}(x)$

**Question 12. The answers for the following questions which require a range of values should be given in set notation.** Given

$$f : x \mapsto 2x^2 + 1$$

for $x \geq 0$.

a) find $y = f^{-1}(x)$

b) show that there are no solutions to $f(x) = f^{-1}(x)$

c) sketch $y = f(x)$ and $y = f^{-1}(x)$ on the same axes

d) state the domain and range of $f^{-1}$

**Question 13.** Given

$$f(x) = 2e^{3x} - 1$$

for $x \in \mathbb{R}$.

a) find $y = f^{-1}(x)$

b) given $f(x) = f^{-1}(x)$ has two solutions, sketch $y = f(x)$ and $y = f^{-1}(x)$ on the same axes

c) state the domain and range of $f^{-1}$

**Question 14.** Given

$$f : x \mapsto 3 + 5^{-x}$$

for $x \in \mathbb{R}$.

a) find $y = f^{-1}(x)$

b) sketch $y = f(x)$ and $y = f^{-1}(x)$ on the same axes

c) state the domain and range of $f^{-1}$

**Question 15.** Given

$$f(x) = \frac{2x}{x+5}$$

for $-4 \leq x \leq 1$.

a) find $y = f^{-1}(x)$

b) solve $f(x) = f^{-1}(x)$

c) sketch $y = f(x)$ and $y = f^{-1}(x)$ on the same axes

d) state the domain and range of $f^{-1}$

**Question 16.** For each of the following functions, find the domain and range of $f^{-1}$:

a) $f(x) = 5x - 2$ for $-1 \leq x \leq 5$

b) $f(x) = \ln x$ for $x > 0$

c) $f(x) = e^{x-2}$ for $x \in \mathbb{R}$

d) $f(x) = \sin x$ for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$

e) $f(x) = \cos x$ for $0 \leq x \leq \pi$

f) $f(x) = \tan x$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$

**Question 17.** A function is given by

$$f(x) = x^4 - x^3 + 2x$$

where $x \in \mathbb{R}$.

a) By using the factor theorem, show that $(x + 1)$ is a factor of $f(x)$

b) Show that $f(x) = 0$ only has two solutions.

c) Explain why an inverse function does not exist.

The domain of $f$ is restricted such that

$$0 \leq x \leq 2.$$

d) Find the domain and range of $f^{-1}$.

**Question 18.** Given

$$f(x) = x^2 - 6x + 5$$

for $x \geq p$ where $p$ is a constant.

a) sketch $y = f(x)$ for $x \in \mathbb{R}$

b) state the minimum value that $p$ can be for $f$ to be one-to-one.

For your value of $p$

c) find the range of $f$

d) state the domain and range of $f^{-1}$

e) show that $f(x) = f^{-1}(x)$ only has one solution.

f) sketch $y = f(x)$ and $y = f^{-1}(x)$ on the same axes.

**Question 19.** Given

$$f(x) = \ln(x - 3)$$

for $x > p$ where $p$ is a constant.

a) State the value of $p$ such that $f$ has the largest possible domain.

For your value of $p$

b) find the range of $f$

c) find $y = f^{-1}(x)$

d) state the domain and range of $f^{-1}$

e) sketch $y = f(x)$ and $y = f^{-1}(x)$ on the same axes.

**Question 20.** Given

$$f(x) = \sqrt{x - 5}$$

for $x \geq p$ where $p$ is a constant.

a) State the value of $p$ such that $f$ has the largest possible domain.

For your value of $p$

b) find the range of $f$

c) find $y = f^{-1}(x)$

d) state the domain and range of $f^{-1}$

It is known that $x^4 + 10x^2 - x + 30 = 0$ has no solutions.

e) Hence, by considering $f(x) = f^{-1}(x)$, show that there is no intersection between the graphs of $y = f(x)$ and $y = f^{-1}(x)$.

f) sketch $y = f(x)$ and $y = f^{-1}(x)$ on the same axes.

**Question 21.** Given

$$f(x) = 3 - \frac{1}{x+2}$$

for $x \neq p$ where $p$ is a constant.

a) state the value of $p$

For $x > p$

b) find the range of $f$

c) find $y = f^{-1}(x)$

d) state the domain and range of $f^{-1}$

e) state the coordinates of the point of intersection of the graphs of $y = f(x)$ and $y = f^{-1}(x)$

f) sketch $y = f(x)$ and $y = f^{-1}(x)$ on the same axes.

**Question 22.** Given

$$f(x) = \frac{4x-1}{x-5}$$

for $x \neq p$ where $p$ is a constant.

a) state the value of $p$

b) find the range of $f$

c) state the domain and range of $f^{-1}$

**Question 23.** For each of the following functions find $f^{-1}(1)$:

a) $f : x \mapsto 5x + 3$

b) $f : x \mapsto x^2 - 8, \quad x \geq 0$

c) $f : x \mapsto \frac{4x+9}{2x+1}, \quad x \neq -\frac{1}{2}$

**Question 24.** For

$$f(x) = ax + b$$

where $a$ and $b$ are positive constants, find

a) the gradient function of $f$

b) the gradient function of $f^{-1}$

c) the values of $a$ such that the gradient of $f(x)$ is equal to the gradient of $f^{-1}(x)$

**Question 25.** For

$$f(x) = x^2, \quad x \geq 0$$

a) find the gradient of $f(x)$ at $x = 2$

b) find the gradient of $f^{-1}(x)$ at $x = 2$

c) find the value of $x$ such that the gradient of $f(x)$ is equal to the gradient of $f^{-1}(x)$

**Question 26.** Given

$$f(x) = -x + 5 \qquad \text{and} \qquad g(x) = 2x - 1$$

for $x \in \mathbb{R}$

a) find $fg(x)$

b) find the value of constant $k$ such that $fg(5) = g^{-1}(k)$

**Question 27.** Given

$$f(x) = 2x^2 - 8x \qquad \text{and} \qquad g(x) = 5x + 1$$

for $x \in \mathbb{R}$

a) find $fg(x)$

b) find the value of constant $k$ such that $fg(2) = g^{-1}(k)$

**Question 28.** Given

$$f(x) = -2x + 1 \qquad \text{and} \qquad g(x) = 3x + 5$$

$x < 0$ for $f$, $\quad x > 1$ for $g$

a) find $gf(x)$

b) find the domain of $gf$

c) state the range of $gf$

d) find $(gf)^{-1}(x)$

e) state the domain and range of $(gf)^{-1}(x)$

**Question 29.** Given

$$f(x) = x + 4 \qquad \text{and} \qquad g(x) = x^2 - x$$

$x < 0$ for $f$, $\quad x > 1$ for $g$

a) find $gf(x)$

b) find the domain of $gf$

c) state the range of $gf$

d) find $(gf)^{-1}(x)$

e) state the domain and range of $(gf)^{-1}(x)$

**Question 30.** Given

$$f(x) = 3x + 1 \qquad \text{and} \qquad g(x) = \frac{2x + 2}{x + 2}$$

$x \neq -2$ for $g$

a) find $fg(x)$

b) find the domain of $fg$

c) state the range of $fg$

d) find $(fg)^{-1}(x)$

e) state the domain and range of $(fg)^{-1}(x)$

**Question 31. The answers for the following questions which require a range of values should be given in interval notation.** Given

$$f(x) = x^2 - 6x + 1$$

for $x \leq 1$.

a) state the range of $f$

b) find $y = f^{-1}(x)$

c) find the solution to $f(x) = f^{-1}(x)$

d) sketch $y = f(x)$ and $y = f^{-1}(x)$ on the same axes

e) state the domain and range of $f^{-1}$

**Question 32.** Given

$$f : x \mapsto e^{2x} + 2$$

for $x \in \mathbb{R}$.

a) find $y = f^{-1}(x)$

b) sketch $y = f(x)$ and $y = f^{-1}(x)$ on the same axes

c) state the domain and range of $f^{-1}$

**Question 33.** Given

$$f(x) = \frac{3x}{x-4}$$

for $-5 < x < 2$.

a) find $y = f^{-1}(x)$

b) solve $f(x) = f^{-1}(x)$

c) sketch $y = f(x)$ and $y = f^{-1}(x)$ on the same axes

d) state the domain and range of $f^{-1}$

**Question 34.** For each of the following functions, find the domain and range of $f^{-1}$:

a) $f(x) = -2x + 3$ for $-2 \leq x \leq 7$

b) $f(x) = 2\ln x$ for $x > 0$

c) $f(x) = 2 + e^{x+5}$ for $x \in \mathbb{R}$

**Question 35.** Given

$$f(x) = \ln(x + 2)$$

for $x > p$ where $p$ is a constant.

a) state the value of $p$

For your value of $p$

b) find the range of $f$

c) find $y = f^{-1}(x)$

d) state the domain and range of $f^{-1}$

e) Given $f(x) = f^{-1}(x)$ has two solutions, sketch $y = f(x)$ and $y = f^{-1}(x)$ on the same axes.

**Question 36.** Given

$$f(x) = \frac{5x + 2}{x - 9}$$

for $x \neq p$ where $p$ is a constant.

a) state the value of $p$

b) find the range of $f$

c) state the domain and range of $f^{-1}$

**Question 37.** Given

$$f : x \mapsto 2x^2 + 4x - 5$$

for $x \geq p$ where $p$ is a constant.

a) state the minimum value that $p$ can be for $f$ to be one-to-one.

For your value of $p$

b) find the domain and range of $f^{-1}$

c) sketch $y = f(x)$ and $y = f^{-1}(x)$ on the same axes.

**Question 38.** Given

$$f(x) = 1 + 2x, \quad x < 1 \qquad \text{and} \qquad g(x) = x^2, \quad x > 2$$

a) find $gf(x)$

b) find the domain of $gf$

c) state the range of $gf$

d) find $(gf)^{-1}(x)$

e) state the domain and range of $(gf)^{-1}(x)$

**Question 39.** Given

$$f(x) = -2x + 3 \qquad \text{and} \qquad g(x) = \frac{8x + 21}{x + 7}$$

a) find $fg(x)$

b) find the domain of $fg$

c) state the range of $fg$

Given $g$ is now restricted such that $x > -7$

d) find $(fg)^{-1}(x)$

e) state the domain and range of $(fg)^{-1}(x)$